

Observer models — step-by-step comparison

A single-view comparison of the observer models built for this project, aligned stage by stage. Each model's own equation labels are given in full in its individual reference:

Model	File	Equation labels
Switch — paper's Switching observer (Laquittaine & Gardner 2018)	switching_observer.py	H1–H5
Basic Bayesian — paper's baseline integrator (Laquittaine & Gardner 2018)	basic_bayesian.py	paper Eq 1–7
HB - Rachel — our hb_integration observer (the abstract's model)	hb_integration.py	M1–M5
HB - Salma — independent hierarchical observer	hierarchical_confidence/model.py	B1–B5
Recombined — Rachel × Salma cross	hb_integrate_before.py	(inherits M1–M5)
HB - Adaptive — the abstract's model taken literally (learns α and κ)	hb_adaptive_confidence.py	(inherits M1–M5)
Rommana — reliability-mixture observer (learned mixture weight)	reliability_mixture_model.py	R1–R5

The models do not number 1:1 (the Switch keeps evidence and prior as two separate read-outs; Basic Bayesian multiplies them into one posterior; the HB models integrate them *and* learn), so the tables below align on **conceptual stages** rather than on each model's internal step number.

Two of the seven are the paper's own **non-learning** models — the **Switch** (prior and evidence kept separate, committed to one or the other per trial) and **Basic Bayesian** (prior and evidence multiplied into a single posterior). The other four are our **learning** models: **HB - Rachel**, **HB - Salma**, **Recombined**, and **HB - Adaptive** all learn a prior latent trial-by-trial. Basic Bayesian is the key foil — "the Switch minus the switch": same nine parameters, but it *integrates* instead of *selecting*, so its estimate is **unimodal**. It is the model the Switch was shown to beat, and the starting point our HB models add learning on top of.

Recombined is defined by one axis only: it is exactly HB - Rachel with the *combination rule* swapped to HB - Salma's (integrate the belief into ONE prior *before* reading out, rather than averaging per- κ read-outs *after*). Every other stage is inherited byte-for-byte from HB - Rachel — same α floor, same learned belief, same linear forgetting, same 360-bin grid, same 7 parameters. Read it as "HB - Rachel's engine with HB - Salma's combination rule."

HB - Adaptive is also defined by one axis relative to HB - Rachel, but a different one: the prior-confidence weight α , **which Rachel fits as a fixed constant, is here learned trial-by-trial** alongside the width κ . The observer carries a **joint belief** $\mathbf{b}_t(\kappa, \alpha)$ over a 2-D grid of (width, confidence) pairs and updates both latents from feedback with Rachel's own predict/correct filter. Everything else — integrate-after combination, linear forgetting, 360-bin grid, motor+lapse, and the underlying `mixture_prior_map_readout` kernels — is inherited from HB - Rachel (it imports them directly). Because α moves from the *parameter budget* into the *latent state*, it has **one fewer free parameter than Rachel** (6 vs 7). Read it as "HB - Rachel with the confidence weight promoted from a fitted constant to a learned latent." It is the only **integrating** model in which *how much to trust the prior* is itself learned — the abstract's literal requirement. (Rommana also learns a trust-the-prior weight, but within a *mixture* architecture, not graded integration — see its framing below.)

Rommana sits apart from the four HB models on the *defining* axis: like the **Switch**, its percept is a genuine **discrete either/or mixture** — a prior-centered component or a likelihood-centered component, never a multiplicative blend — so it does not integrate. But unlike the Switch, whose mixture weight is *derived* from the reliability ratio and fixed per block, Rommana **learns its mixture weight prior_reliance trial-by-trial** with a delta rule: after each trial the weight moves toward how well recent (smoothed) feedback agrees with the 225° prior mean, at learning rate α . So it is the one model that keeps the Switch's *architecture* (select, don't integrate) while adding the HB models' *learning* (the reliance on the prior is acquired from feedback, not set by hand). Read it as "the Switch, but the coin's bias is learned." What it learns is the *weight* (how much to trust the prior), whereas the HB models learn the prior's *width* κ ; it does not marginalise over a κ belief and has no MAP read-out — the mixture density itself is the response distribution. It is committed on `main` (Rommana's `reliability_mixture_hb_model/`).

Stage-by-stage comparison

Stage 1 — Prior representation

Aspect	Switch	Basic Bayesian	HB - Rachel	HB - Salma	Recombined	HB - Adaptive	Rommana
Form	Delta at 225°	Pure $V(\theta; 225, \kappa_{\text{prior}})$	$\alpha \cdot V(\theta; 225, \kappa) + (1 - \alpha) / 360$	Pure $V(\theta; 225, \kappa)$, per κ	$\alpha \cdot V(\theta; 225, \kappa) + (1 - \alpha) / 360$	$\alpha \cdot V(\theta; 225, \kappa) + (1 - \alpha) / 360$, per (κ, α) pair	$V(\theta; 225, \kappa_{\text{prior}})$ — the prior-centered mixture component
Mixture floor (uniform)	via lapse only	via lapse only	yes, weight $(1 - \alpha)$	no ($\kappa=0$ grid point)	yes, weight $(1 - \alpha)$	yes, $(1 - \alpha)$ — and α is learned	via lapse only
Prior width set by	fitted <code>k_prior/block</code>	fitted <code>k_prior/block</code>	learned belief <code>b_t(κ)</code>	learned belief <code>H_t(κ)</code>	learned belief <code>b_t(κ)</code>	joint learned belief $\mathbf{b}_t(\kappa, \alpha)$	fitted <code>k_prior/block</code> (width fixed; <i>weight</i> is learned)
Label	H2	Eq 2	M1	B1	M1 (inherited)	M1 (inherited)	R1

Stage 2 — Sensory / likelihood component

Aspect	Switch	Basic Bayesian	HB - Rachel	HB - Salma	Recombined	HB - Adaptive	Rommana
Form	$V(\cdot; \theta_{\text{true}}, \kappa_{\text{like}}(c))$	$V(\cdot; \theta_{\text{true}}, \kappa_{\text{like}}(c))$	$V(\cdot; \theta_{\text{true}}, \kappa_{\text{like}}(c))$	$V(\cdot; \theta_{\text{true}}, \kappa_{\text{sensory}}(c))$	$V(\cdot; \theta_{\text{true}}, \kappa_{\text{like}}(c))$	$V(\cdot; \theta_{\text{true}}, \kappa_{\text{like}}(c))$	$V(\cdot; \theta_{\text{true}}, \kappa_{\text{llh}}(c))$ — likelihood-centered mixture component
Measurement noise	marginalised (Girshick)	marginalised (Girshick)	marginalised (Girshick)	marginalised (Girshick)	marginalised (Girshick)	marginalised (Girshick)	not marginalised — VM centered on true direction
Per-coherence reliability	<code>3 k_like</code>	<code>3 k_like</code>	<code>3 k_like</code>	<code>3 k_sensory</code>	<code>3 k_like</code>	<code>3 k_like</code>	<code>3 k_llh</code>
Label	H1	Eq 1	M2	B2 (grid)	M2 (inherited)	M2 (inherited)	R2

Stage 3 — How prior and evidence are combined (the defining axis)

Aspect	Switch	Basic Bayesian	HB - Rachel	HB - Salma	Recombined	HB - Adaptive	Rommana
Rule	MIXTURE — commit per trial	INTEGRATE — one posterior	INTEGRATE, avg read-outs	INTEGRATE into one prior	INTEGRATE into one prior	INTEGRATE, avg read-outs	MIXTURE — weighted either/or

Formula	$P_e\text{-evid} + P_{pr}\text{-prior} + P_r\text{-unif}$	$MAP[V(\theta_e, k_e) \cdot V(225, k_{prior})]$	$\Sigma_{\kappa} b(\kappa) \cdot R_{\kappa}$ (avg after)	$MAP(\Sigma_{\kappa} H(\kappa) \cdot V)$ (one before)	$MAP(\Sigma_{\kappa} b(\kappa) \cdot V)$ (one before)	$\Sigma_{(\kappa, \alpha)} b(\kappa, \alpha) \cdot R_{(\kappa, \alpha)}$ (avg after)	$r\text{-prior} + (1-r) \cdot llh$, r = learned reliance
κ marginalised...	n/a	n/a	AFTER read-out	BEFORE read-out	BEFORE read-out	AFTER read-out (over κ and α)	n/a (no κ belief)
Bimodality is...	imposed by commit	none — unimodal	emergent (floor)	emergent (posterior)	emergent (floor)	emergent (floor)	imposed by mixture
Label	H3+H4	Eq 3–5	M4	B3+B4	B3-style + M's α floor	M4 (inherited)	R3

Stage 4 — Reliance-on-prior weight

Aspect	Switch	Basic Bayesian	HB - Rachel	HB - Salma	Recombined	HB - Adaptive	Rommana
What sets it	derived $k_{prior}/(k_{prior}+k_e)$	emergent (precision-weight)	fitted α (fixed)	emergent from belief	fitted α (fixed)	learned α (latent, tracked)	learned reliance r (latent, tracked)
Free parameter?	no	no	yes (α)	no	yes (α)	no — α is a learned latent	no — r is a learned latent (α is its learning <i>rate</i>)
Label	H3	Eq 3–5	M1	—	M1 (inherited)	M1 (inherited)	R4

Stage 5 — Learned latent (trial-by-trial dynamics) ← where the HB models sit

Aspect	Switch	Basic Bayesian	HB - Rachel	HB - Salma	Recombined	HB - Adaptive	Rommana
Learns across trials?	no (fixed/block)	no (fixed/block)	yes — $b_t(\kappa)$	yes — $H_t(\kappa)$	yes — $b_t(\kappa)$	yes — $b_t(\kappa, \alpha)$	yes — reliance r_t (scalar)
WHAT is learned	—	—	prior concentration κ	prior concentration κ	prior concentration κ	prior concentration κ AND confidence α	prior-reliance weight r (mixture weight)
Update rule	none	none	forget → Bayes-correct	forget → Bayes-correct	forget → Bayes-correct	forget → Bayes-correct (joint)	delta rule $r \leftarrow r + \alpha \cdot (\text{agreement} - r)$
Forgetting form	—	—	linear $(1-\lambda)b+\lambda \cdot b_0$	geometric $\log H \leftarrow p \cdot \log H$	linear $(1-\lambda)b+\lambda \cdot b_0$	linear $(1-\lambda)b+\lambda \cdot b_0$	EMA over feedback window (no explicit b_0)
Reset	n/a	n/a	no (carried)	no (carried)	no (carried)	no (carried)	yes — at session boundaries
Label	—	—	M5	B2	M5 (inherited)	M5 (inherited, over $\kappa \times \alpha$)	R5

Stage 6 — Read-out type

Aspect	Switch	Basic Bayesian	HB - Rachel	HB - Salma	Recombined	HB - Adaptive	Rommana
Read-out	MAP (Girshick)	MAP on joint posterior (once)	MAP per κ (Girshick)	MAP, tie-aware	MAP on effective prior (once)	MAP per (κ, α) pair (Girshick)	none — mixture density is the response pmf
Tie handling	argmax	argmax	argmax	ties share 1/#ties	argmax	argmax	n/a (no argmax)
Label	H1/H2	Eq 3–5	M3	B4	M3 (inherited)	M3 (inherited)	R3

Stage 7 — Motor noise + lapse

Aspect	Switch	Basic Bayesian	HB - Rachel	HB - Salma	Recombined	HB - Adaptive	Rommana
Motor	■ $V(\cdot; 0, k_{motor})$	■ $V(\cdot; 0, k_{motor})$	■ $V(\cdot; 0, k_{motor})$	■ $V(\cdot; 0, k_{motor})$ FFT	■ $V(\cdot; 0, k_{motor})$	■ $V(\cdot; 0, k_{motor})$	■ $V(\cdot; 0, k_{motor})$
Lapse	into Eq. 6 weights	$(p+p_r\text{-unif})/(1+p_r)$	$(p+p_r\text{-unif})/(1+p_r)$	$(1-\text{lapse}) \cdot p + \text{lapse}/n_{\theta}$	$(p+p_r\text{-unif})/(1+p_r)$	$(p+p_r\text{-unif})/(1+p_r)$	$(1-\text{lapse}) \cdot p + \text{lapse}/360$
Label	H5	Eq 7	M4	B5	M4 (inherited)	M4 (inherited)	R5

Numerical grid & fitting

Aspect	Switch	Basic Bayesian	HB - Rachel	HB - Salma	Recombined	HB - Adaptive	Rommana
Direction grid	360 (1°)	360 (1°)	360 (1°)	72 (5°)	360 (1°)	360 (1°)	360 (1°)
κ grid	n/a	n/a	~15 points	16 points (incl. $\kappa=0$)	~15 points	~15 $\kappa \times 9 \alpha = 135$ pairs	n/a (no κ belief; scalar reliance)
Compute	linear	linear	linear	log (logsumexp)	linear	linear	linear
Optimiser	(fitter-dependent)	(fitter-dependent)	(fitter-dependent)	Powell + LH multistart	(fitter-dependent)	(fitter-dependent)	(fitter-dependent)

Parameter counts

Model	N	Parameters
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Switch	9	$3 k_like + 4 k_prior + k_motor + p_random$
Basic Bayesian	9	$3 k_like + 4 k_prior + k_motor + p_random$
HB - Rachel	7	$3 k_like + \alpha + k_motor + p_random + \lambda$
HB - Salma	6	$3 k_sensory + p + k_motor + lapse$
Recombined	7	$3 k_like + \alpha + k_motor + p_random + \lambda$
HB - Adaptive	6	$3 k_like + k_motor + p_random + \lambda$ (α and κ are learned latents, not fitted)
Rommana	10	$3 k_lh + 4 k_prior + \alpha$ (reliance learning rate) + $k_motor + lapse$ (reliance r is a learned latent)

Switch and **Basic Bayesian** share the **identical nine-parameter set** — the only difference between them is Stage 3 (select vs multiply), exactly the paper's comparison. **Recombined** and **HB - Rachel** share the **identical seven-parameter set** — the only difference between *them* is also Stage 3 (average per- κ read-outs after vs integrate into one prior before). The combination rule is thus the axis that separates *both* matched-parameter pairs. **HB - Adaptive** and **HB - Rachel** share the **identical combination rule and engine** — the only difference between *them* is Stage 4/5: Rachel *fits* α as a fixed constant, while Adaptive *learns* it as a latent, dropping it from the parameter budget (7 $\rightarrow$ 6). α **appears explicitly in HB - Rachel and Recombined** as the fixed fitted prior mixture weight; in **HB - Adaptive** α exists but is learned, not fitted; the remaining two models have no α .

One-line summary of the differences

- **The two paper baselines (Switch, Basic Bayesian) both fix the prior width** per block — neither learns trial-by-trial. They differ only in how they combine prior and evidence: the **Switch selects** (commits per trial $\rightarrow$ the **bimodal** estimates subjects showed), while **Basic Bayesian multiplies** (one posterior $\rightarrow$ a single **unimodal** bump). Basic Bayesian is the model the Switch was shown to beat.
- **The four HB models all add learning** on top of integration: they learn a prior latent trial-by-trial from feedback instead of fixing it per block. They differ along two axes:
- **Combination rule (Stage 3):** HB - Rachel and **HB - Adaptive** average per- κ read-outs *after*; HB - Salma and **Recombined** integrate the belief into *one* prior *before* reading out.
- **What is learned (Stage 5):** HB - Rachel, HB - Salma, and Recombined learn only the prior *width* κ (α is fixed or absent). **HB - Adaptive** learns the prior *confidence* α **as well as** κ , via a joint belief $b_l(\kappa, \alpha)$ — the only model in which *how much to trust the prior* is itself a learned latent.
- **Everything else (α floor, forgetting form, grid):** HB - Rachel, **Recombined**, and **HB - Adaptive** share the α -mixture floor, linear forgetting, and 360-bin grid; HB - Salma differs (no α , geometric forgetting, 72-bin log-space).
- So two "one-axis-from-Rachel" crosses sit either side of her: **Recombined = HB - Rachel's engine + HB - Salma's combination rule** (isolates integrate-before vs integrate-after), and **HB - Adaptive = HB - Rachel with α promoted from a fitted constant to a learned latent** (isolates fitting-confidence vs learning-confidence).
- **Rommana is the bridge between the two families.** It keeps the **Switch's architecture** — a discrete either/or mixture that commits to prior *or* evidence per trial (so its bimodality is *imposed*, like the Switch's, not *emergent* like the HB models') — but adds the **HB models' learning**: the mixture weight (reliance on the prior) is acquired trial-by-trial from how well feedback agrees with the prior mean, via a delta rule, and reset each session. It is the only model that learns the *weight* (how much to trust the prior) rather than the prior's *width* κ . Read it as "the Switch with a learned coin bias." It carries the most free parameters (10) because it keeps the Switch's full 3+4 concentration set *and* adds a reliance learning rate.
- **The through-line:** Basic Bayesian (integrate, no learning, unimodal) is the baseline; the Switch (select, no learning) explains the bimodality the baseline misses; the HB models (integrate *with* learning) recover that bimodality *without* a switch rule — as an emergent property of a learned, mixture-floored prior. HB - Rachel/Salma/Recombined explain how prior *width* is acquired; **HB - Adaptive** goes one step further and learns the prior *confidence* too, matching the abstract's literal claim that the observer learns how much to trust its prior.

(Empirical note: HB - Rachel and Recombined agree near the prior and diverge in the far-from-prior bimodality regime — the integrate-before rule leaves faint discretisation ripple that the average-after rule washes out; mean total-variation between the two = 0.16 far, = 0.06 near. HB - Adaptive shares Rachel's integrate-after read-out, so it inherits her far-band shape; its distinctive behaviour is dynamic — $E[\alpha]$ rises toward 1 when feedback clusters near 225° and falls toward 0 when feedback is spread, verified in a 9/9 test battery: bit-exact reduction to a single mixed-prior read-out when the belief is pinned, correct confidence-learning direction, and clean recovery of its six fitted parameters.)